FIT3179 Data Visualisation

Course Notes

MAP PROJECTIONS

7 September 2026

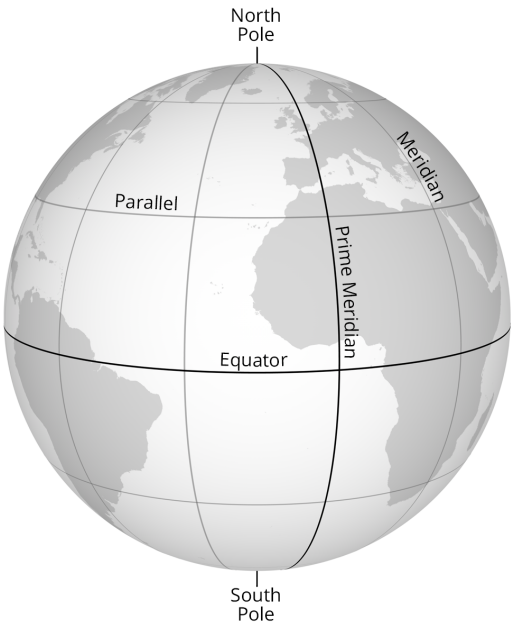
SPHERICAL COORDINATES

The rotation axis of the Earth defines the location of the North Pole and the South Pole. Half-circles on the sphere running from the North Pole to the South Pole are called meridians. There are an infinite number of meridians. The globe on this page shows meridians with a spacing of 30°. The Prime Meridian passes through Greenwich, in London and is the origin for longitudinal angles. Longitude is measured relative to the Prime Meridian and is positive eastwards and negative westwards. Its range is between –180° and +180°.

The equator is the circle around the sphere halfway between the two poles that divides the sphere into two hemispheres. The equator is the origin for latitudinal angles. Latitude is measured starting at the equator and is positive towards the North Pole and negative towards the South Pole. Its range is between –90° and +90°.

A parallel is a circle of constant latitude, which is parallel to the equator, hence the name. As with meridians, there exist an infinite number of parallels. The globe on this page has parallels drawn for every 30°.

Earth is a three-dimensional object, but when we assume that Earth is a perfect sphere, we need only a two-dimensional coordinate system (latitude and longitude) to define any location on its surface.



MAP PROJECTION DEFINITION

A map projection is a transformation from the curved two-dimensional surface of a globe to a plane.

A map projection is normally defined by two equations that convert spherical longitude/latitude coordinates to Cartesian XY coordinates.

SIZE OF EARTH

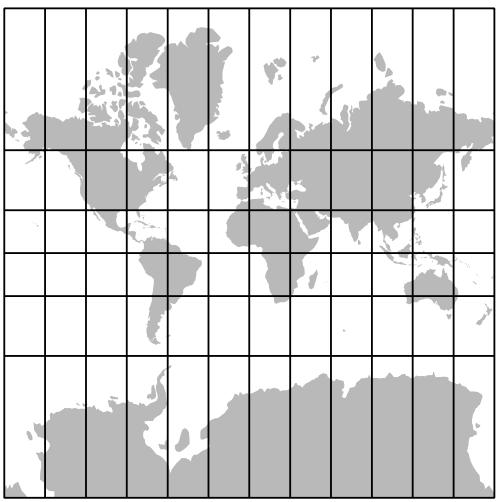
The size of Earth is simple to remember. The metre unit was originally defined by stating that the distance between the equator and one of the poles is 10,000 km. The full circumference of the globe is therefore approximately 40,000 km. The radius is 6,371 km.

MAP PROJECTION DISTORTION

Transforming the curved globe surface to a plane is impossible without distortion. Understanding distortion is important when selecting a projection for a map.

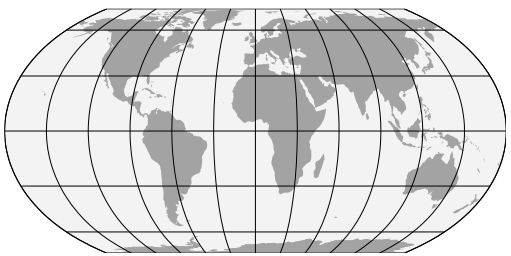
There exist two fundamental types of distortion introduced by map projections: areal distortion and angular distortion.

The Mercator projection is an example that distorts area (below). Areas are increasingly exaggerated towards the poles, which is why Antarctica appears enormous For this reason, the Mercator projection is not a good choice for a world map. The Mercator projection, however, does not distort angles, which makes it a useful projection when angles are measured on a map, for example for navigation at sea.



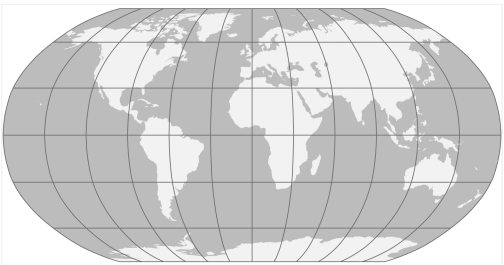
Equal Earth is an example of an equal-area map projection (below). Equal-area projections do not distort the relative size of areas. For example, Greenland is 14 times smaller in area than Africa. On a map using an equal-area projection, this ratio is preserved: On the map, Greenland is also 14 times smaller than Africa. However, this is not the case on a map using a projection that does not preserve relative areas. For example, on a map using the Mercator projection, Greenland is greatly inflated in size because it is close to a pole, resulting in Greenland and Africa being shown at a similar size.

A disadvantage of equal-area projections is that they distort the shape of landforms. In general, shapes are more distorted when they are closer to the border of the map. For example, on the Equal Earth map, New Zealand is rotated and Alaska is distorted.

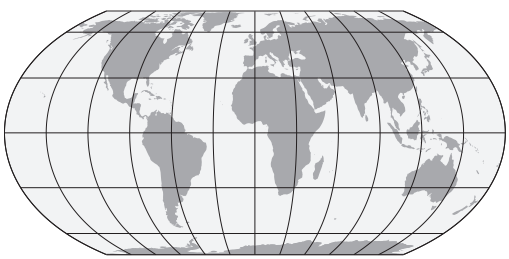


PROJECTIONS FOR WORLD MAPS

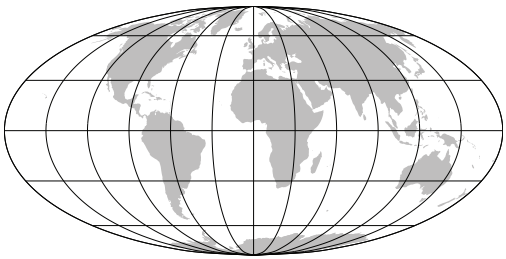
Many projections for world maps exist. Such projections either are equal-area projections that do not distort area, or compromise projections that accept a small amount of area distortion to show landforms with less deformation.



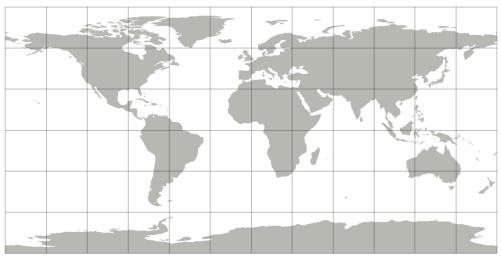
The Robinson projection (above) is a compromise projection that distorts both areas and angles. It slightly enlarges area towards the poles to more clearly show their shapes. The Robinson projection is widely used, and many people consider it visually pleasing.



The Equal Earth projection (above) is an equal-area projection. The Equal Earth map projection is very similar to the Robinson projection, but areas near the poles are not enlarged. Compare, for example, the size of Antarctica on both maps.



The Mollweide projection also does not distort areas. The outer shape is an ellipse, which hints at the spherical shape of Earth. However, the shape of landforms is considerably deformed near the poles.



The equirectangular projection transforms meridians and parallels to straight lines of constant spacing. It distorts areas and angles, and is often used to distribute geodata.

PROJECTIONWIZARD.ORG

ProjectionWizard.org simplifies the selection of map projections for world and regional maps. The user selects the area shown on the map and the property that you want to preserve, and ProjectionWizard suggests an appropriate projection.

PROJECTIONS FOR REGIONAL MAPS

When selecting a projection for a map that shows a section of Earth, such as a continent, a country, or a smaller area, there are three classical types of map projections to choose from. They are grouped by their developable surface. A developable surface is a geometric shape that can be unwrapped to a flat plane. There are three developable surfaces: cylinders, cones, and planes.

Conceptually, a map projection for a regional area transforms coordinates from the sphere to these developable surfaces, which are then flattened (or *developed*) to create the final map. In practice, only a few projection equations use developable surfaces, but the concept of developable surface is nevertheless useful for selecting a map projection that minimises distortion.

The resulting groups of projections are called cylindrical projections, conic projections, and azimuthal (or planar) projections.

When selecting a projection for a regional map, the first step is to identify the aspect ratio of the mapped area: (1) landscape, (2) portrait, or (3) square.

The second step is to identify the central latitude of the area: (1) close to the equator, (2) between the equator and a pole, or (3) at a pole.

With these two pieces of information, the developable surface that most tightly fits the mapped area on the globe can be selected.

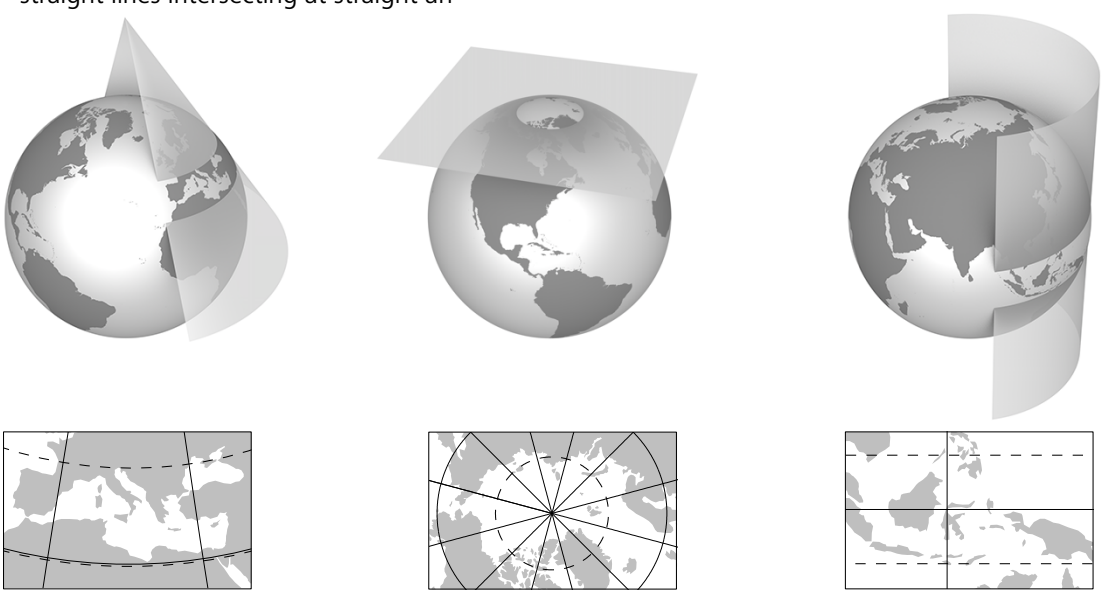
The developable surfaces below illustrate three examples. When the area shown on the map has a landscape aspect ratio and it is located along the equator, then a cylinder fits the area tightly, and a cylindrical map projection is chosen (left). When the area has a landscape aspect ratio and is located between the equator and a pole, then a cone fits the area tightly, and a conic map projection is chosen (centre). When the area has a square aspect ratio, a plane fits the area most closely, and an azimuthal (or planar) map projection is used (right).

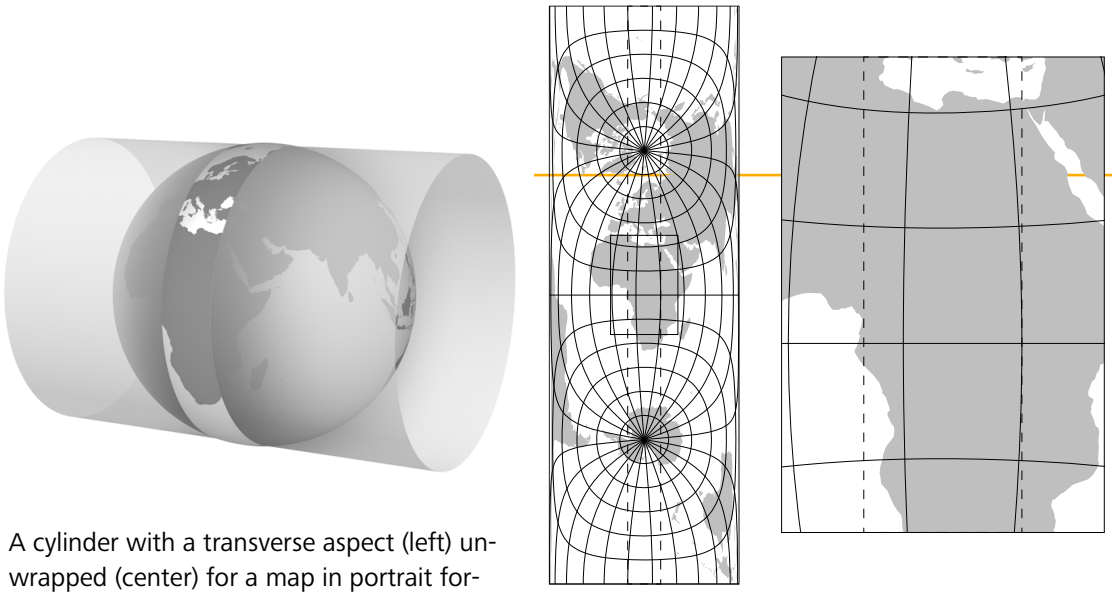


PROJECTION TYPES

The many map projections can be grouped into five categories: the three developable surfaces, plus a category for projections for world maps and one for special cases.

- **Conic** projections are useful for showing a section of the globe on a map with a landscape orientation. They show lines of constant longitude as straight lines and lines of constant latitude as arcs of circles.
- **Azimuthal or planar** projections are useful for square-format maps, for showing the poles with a spider-web-like pattern of meridians and parallels of latitude, and for showing a hemisphere.
- **Cylindrical** projections are useful for maps showing an area along the equator. They show parallels of latitude and meridians as straight lines intersecting at straight angles. Cylindrical projections can also be used with a so-called transverse aspect to show an area with a portrait format. Transverse cylindrical projections have curved parallels and meridians.
- **Pseudocylindrical** projections show parallels of latitude as straight, parallel horizontal lines, while meridians are curved (except for the straight central meridian). Most pseudocylindrical projections show the poles as lines to reduce the distortion of shapes in polar areas. The Robinson, Equal Earth, and Mollweide map projections are examples of pseudocylindrical projections. Pseudocylindrical projections are only useful for world maps.
- Many **other** projections exist that do not fit neatly into the four categories above.





A cylinder with a transverse aspect (left) unwrapped (center) for a map in portrait format (right).



The Spilhaus map projection is an example that belongs to the *other* category. It was designed to show all of Earth's oceans as a connected area, which is useful for mapping the seafloor.

WEB MERCATOR

Tableau, Google Maps, OpenStreetMap, and many other mapping tools use the Web Mercator projection, which is a variation of the Mercator projection with a square format. The Mercator projection is appropriate for maps that zoom in on a small area because then the area distortion across the map is not large. However, it is a poor choice for world maps because it dramatically inflates the size of polar areas. The figure below shows a map with the Mercator projection (bright blue) with countries scaled to their correct relative size overlaid (dark blue).



The Mercator projection is a cylindrical projection that shows parallels and meridians as straight lines. This is very convenient for developers of web maps because north is always oriented upwards, which is not the case with many other projections.



The Web Mercator projection also clips the map at latitudes of about ±85°, resulting in a square map. A square map is convenient to subdivide into regular tiles at a series of zoom levels, and a tiled image pyramid.

Zoom level 0 consists of a single square tile of 256×256 pixels showing the entire globe within ±85° latitude. Zoom level 1 doubles the number of tiles in both directions and uses four tiles. The level 1 map is twice as large and contains more detail. The same tiling scheme is used for level 2 and all subsequent levels.